

Nikhitha Joy

Data Scientist | Machine Learning Developer | Generative AI

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OVERVIEW

AI and Data Science Engineer with over 1 year of professional experience in designing, building, and optimizing intelligent software applications. Proven track record in full-cycle development of machine learning and data-driven systems with clean coding practices, version control, debugging, and testing. Adept in collaborating with cross-functional teams and clients to deliver scalable, high-performance AI services and software components. Demonstrates strong communication, documentation, and continuous improvement mindset aligned with enterprise-grade software quality.

SKILLS

Languages: Python, SQL, R, HTML, CSS, Bash

Frameworks & Libraries: scikit-learn, NumPy, Pandas, TensorFlow, PyTorch, Keras, OpenCV, Transformers, LangChain, LlamaIndex, SpaCy, NLTK, Matplotlib, Seaborn

ML & AI: Supervised & Unsupervised Learning, Deep Learning, Time-Series Forecasting, NLP, Generative AI, LLMs, Open Source LLMs, RAG, LoRA Fine-Tuning, Embedding Models, Semantic Search, Few-shot learning

Cloud & Platforms: Azure ML Studio, Docker, REST APIs, GitHub Actions, CI/CD, Streamlit

Tools & Concepts: Power BI, Vector DB, Prompt Engineering, MLOps, Linux (Microservices), Vector Databases, Web Scraping, RDBMS, ORM, Data Warehousing, ETL Pipelines,

Soft Skills: Problem Solving, Communication, Leadership, Critical Thinking, Cross-functional Collaboration, Agile methodology, technical documentation

EXPERIENCE

Machine Learning Developer

March 2025 - Present

Kofuku Idea Labs, Hyderabad

- Led projects for clients including Telco, Tenneco.
- Designed semantic search systems using LLM models for enterprise knowledge extraction, increasing accuracy by 65% and enhancing processing speed by 40%.
- Developed prompt-optimized AI pipelines to reduce token usage and inference costs.
- Containerized model inference using Docker and integrated FastAPI endpoints with scalable CI/CD deployment workflows.
- Collaborated with product and backend teams to deliver end-to-end AI capabilities in manufacturing use cases.

Data Science and AI Engineer

July 2024 - February 2025

Intelance, London

- Led the Digital Healthcare Platform project, focusing on model development, deployment, and optimization using Azure ML Studio.
- Developed predictive models for chronic disease risk, achieving 78-97% accuracy and precision.
- Designed strategy for LLM-based predictive analytics in clinical settings, leveraging domain-specific embeddings for survival prediction. Documented workflows with architecture diagrams and reproducible scripts for deployment readiness.
- Built automated model evaluation pipelines using Azure ML Studio with support for versioning and registry.

PROJECTS

Market Share AI Agent for Structured Data Analysis

Developed an autonomous agent to analyze multiple Excel/CSV files and compute product-specific market share over selected time windows. Integrated data ingestion, preprocessing, and computation logic into a single workflow to support client-side deployment.

Business Name Clustering Using LLM + Fuzzy Matching

Built an intelligent clustering system that grouped business names (e.g., Walmart Inc., WAL-MART) using hybrid fuzzy matching and LLM-enhanced contextual understanding. Applied weighted similarity scoring and LLM-based disambiguation to improve cluster precision on large-scale datasets.

EDUCATION

BTech in Artificial Intelligence and Data Science

November 2020 - June 2024

Jyothi Engineering College, Thrissur

CGPA: 7.88/10 (78.8%)

CERTIFICATIONS

Google Project Management Professional Certificate — In Progress